

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460330.737 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.027 +/- 0.048
Transit depth (Rp/Rs)^2	0.073 +/- 0.26 [%]
Orbital Inclination (inc)	86.17 +/- 2.04
Airmass coefficient 1 (a1)	1.01637 +/- 0.00059
Airmass coefficient 2 (a2)	-0.01 +/- 0.29
Scatter in the residuals of the lightcurve fit is	0.002 %
Best Comparison Star	#2 - [268, 32]
Optimal Aperture	2.18
Optimal Annulus	8.81
Transit Duration (day)	0.038 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.027 ± 0.048 vs 0.1241 (deviation 2.02σ)
Duration fit vs published	0.038 d vs 0.1317 d (deviation 6.69σ)
Mid-time O–C	-5.7 min
Depth SNR	0.6
Residual scatter	0.002 %

Light curve

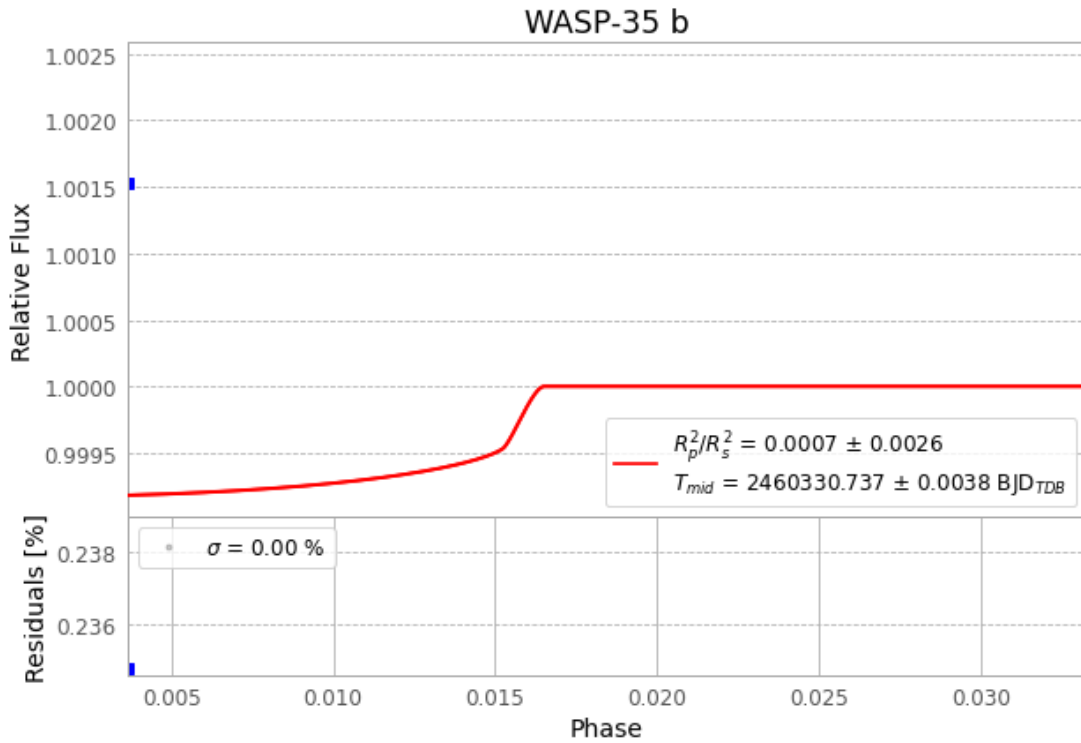


Figure 1 — FinalLightCurve_WASP-35 b_2024-01-20.png (SHA-256 e26603222fea89a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [468, 190], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 befb24b555b12008...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), WASP-35240121041815.FITS → WASP-35240121081816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-240121.log (9a10c51e1d85...), FinalLightCurve_WASP-35 b_2024-01-20.png (e26603222fea...), FinalLightCurve_WASP-35 b_2024-01-20.csv (ea3cc0a6da7e...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 06:59Z.